PARENT-CENTRIC WEB APPLICATION FOR TODDLER DEVELOPMENT MONITORING USING DATA MINING.

Research Project Proposal
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by

Dilupshi NPS

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Table of Contents

1. Introduction 3
1.1 Background of the Study 3
1.2 Motivation 3
1.3 Study Objectives 4
2. Literature Review and Research Gap 5
2.1 Overview of Existing Studies 5
2.2 Identification of Research Gap 5
3. Research Problem and Questions 6
3.1 Main Research Problem 6
3.2 Specific Research Questions 6
4. Research Strategy and Methodology 6
4.1 Research Strategy 6
4.2 Research Methodology Framework 6
4.3 Data Collection Methods 8
4.4 Data Preprocessing / Preparation 8
4.5 Model Development / Implementation 9
4.6 Evaluation 9
4.7 Ethical considerations 9
4.8 Limitations of the study 10
5.Timeline 11
6. References (IEEE Style) 11

1. Introduction

1.1 Background of the Study

The developmental window between birth and three years of age is the most critical period for human brain growth and the establishment of neuro-developmental foundations. During this phase, toddlers acquire complex linguistic, motor, and social-emotional skills that determine their future academic and psychological trajectories. Research suggests that high-quality, center-based early childhood education and proactive family engagement are the most effective ways to alter the school trajectories of children who may be vulnerable due to environmental factors or developmental delays [1][2]

In the current technological landscape, children are being introduced to digital devices at an increasingly younger age. The emergence of "Digitods" - babies and toddlers who interact with touchscreens - has transformed the traditional Australian family home, creating new dynamics in how children play and learn [3]. While these devices offer a form of digital immersion, experts are concerned about the long-term impact on social bonding and the potential for digital addiction if screen time replaces active physical play and face-to-face interaction with caregivers.

1.2 Motivation

The primary motivation for this research is the critical need to protect toddlers from the adverse effects of excessive screen exposure while still leveraging technology for health monitoring. Most current applications require the child to be the primary user, which contradicts clinical advice to limit digital consumption during early infancy [3]. Furthermore, evidence shows that a mother's belief in her own parenting abilities - known as maternal self-efficacy - is a vital mediator in a child's linguistic success [2]. By creating a system that focuses on the parent, we can improve these outcomes by providing them with better knowledge and monitoring tools.

Additionally, early diagnosis of developmental disorders such as Autism is often hindered by high costs and the complexity of professional assessment [13]. If a child’s developmental phenotype can be identified as early as 12 months with stability, it allows for interventions that can significantly improve the child’s quality of life [6]. This project is motivated by the potential of using AI to bridge this gap, providing an accessible, cost-effective screening tool that utilizes the parent's daily observations to identify early warning signs without requiring the child to look at a screen.

1.3 Study Objectives

Primary Objective

1. The primary focus of this research is designing, developing, and evaluating a parent-mediated responsive web application that utilizes behavioral data mining to provide an automated screening of the developmental progress of toddlers between 1 and 3 years old.

Secondary Objectives

The following are the sub-objectives set out to attain the primary objective:

- Guided Physical Activities: To design a repository of age-appropriate, offline developmental activities that parents can perform with their toddlers, promoting milestones in cognition, motor skills, and language without direct screen use.
- Data Acquisition of Interaction: To implement a secure logging system that captures parent-reported interaction data, including milestone success rates and behavioral observations, to reflect the child's true developmental status [11].
- Parent-Centric Design (HCI): To develop an intuitive and professional dashboard interface specifically optimized for parents and caregivers to facilitate easy data entry and progress monitoring [4].
- Data Mining Implementation: To apply advanced algorithms, such as Hybrid Random Forest, to the parental observation logs to identify developmental patterns, outliers, and potential risks [5].
- Personalized Feedback Loop: To design an automated reporting engine that converts data insights into personalized progress reports, offering parents actionable advice benchmarked against international development standards [2][5], [6]

2. Literature Review and Research Gap

2.1 Overview of Existing Studies

Extensive research has been conducted on the role of technology in early intervention and monitoring. For instance, web-based support systems have been successfully used to predict poor performance in students by employing Educational Data Mining (EDM) techniques, demonstrating that predictive models can handle complex behavioral inputs [4]. Furthermore, clinical studies have utilized longitudinal MRI data to track cortical development in children with Autism, proving that brain growth trajectories are identifiable patterns that can be modeled digitally [10].

In the realm of parenting, Albarran and Reich [2] demonstrated that providing mothers with educational content through "Baby Books" significantly increases their self-efficacy and the child's language development. While some researchers have explored the use of social robots for education to enhance cognitive and affective outcomes [9], the consensus remains that technology should support rather than replace human caregivers. These studies collectively provide a foundation for building a system that integrates behavioral science with data mining to monitor growth effectively [5].

2.2 Identification of Research Gap

Despite the abundance of "baby tracking" apps, most current solutions are Android-based tools focused on passive record-keeping or direct child interaction [11]. There is a significant research gap in the development of a web-based Software Engineering artifact that specifically targets the reduction of toddler screen time while performing AI-driven developmental screening. Most existing software does not offer a predictive analysis of social-emotional milestones through the lens of parental observation.

Furthermore, there is a lack of integration between Hybrid Machine Learning models and parent-centric web dashboards in the Sri Lankan context. Most systems provide generic information but fail to offer personalized feedback based on the specific "interaction quality" between parent and child [12]. This project addresses this gap by creating an integrated MERN stack application that prioritizes the parent's observational data as the primary feature set for a Hybrid Random Forest classifier.

3. Research Problem and Questions

3.1 Main Research Problem

The central problem this research addresses is the absence of an accessible, parent-mediated screening tool that can accurately monitor a toddler's developmental progress without exacerbating digital screen addiction. Current assessment methods are either high-cost clinical evaluations or direct-interaction apps that increase the child's exposure to touchscreens [3][13]. There is a critical need for a software solution that utilizes the parent's observations to provide early risk detection through advanced data mining.

3.2 Specific Research Questions

1. How can subjective parental logs regarding motor and linguistic milestones be transformed into objective feature sets that yield high predictive accuracy in a Hybrid Random Forest model [5][7]?
2. To what extent does a parent-centric web dashboard, which provides guided offline activities, increase the maternal self-efficacy and active interaction time between the parent and the toddler [2]?
3. Can the proposed web application effectively identify the early developmental phenotype of disorders like Autism with a degree of accuracy comparable to existing digital screening tools [8][10]?

4. Research Strategy and Methodology

4.1 Research Strategy

This research adopts a Design Science Research (DSR) approach to develop and evaluate a parent-mediated software artifact for toddler developmental assessment. By bridging the gap between clinical observation and automated data mining, this strategy provides a functional solution to reduce excessive screen exposure. The framework ensures the web application is iteratively refined based on behavioral theory and technical performance, resulting in a robust tool for domestic milestone monitoring [1][5].

4.2 Research Methodology Framework

The research follows the DSRM framework proposed by Peffers et al., designed for developing and evaluating innovative IT solutions. Such a framework ensures that the development of a web application is firmly based on theoretical knowledge and practical evaluation. The stages of the methodology are described as follows:

1. Problem Identification and Motivation: The lack of real-time, parent-mediated developmental assessment tools for toddlers was identified as a critical gap. The motivation stems from the urge to convert passive, harmful screen time into meaningful

"offline" cognitive engagement between the parent and child [3]. This phase is informed by an extensive literature review on childhood intervention and Educational Data Mining (EDM) [4][5][13].

2. Define Objectives for the Solution: During this phase, the requirements of the "Responsive Web Application" are determined. The objectives are to design guided, age-appropriate activities for 1–3-year-olds, ensure mobile responsiveness via the MERN stack, and establish parental logging protocols that provide high-quality data for machine learning analysis [2][11].
3. Design and Development: The heart of the research is the development of software artefacts. It involves:
    - Front-end: Creating a parent-friendly UI/UX using **React.js** for easy activity tracking and report viewing.
    - Back-end: A robust server-side implementation using **Node.js** and **Express.js**, with a **MongoDB** database for secure interaction logging.
    - Data Mining Module: A Python-based service applying algorithms like Random Forest or Hybrid models to classify the collected parental logs [5] [8]
4. Demonstration: The prototype is demonstrated through pilot sessions where parents interact with the system. At this stage, parents log their child's success in specific tasks under guided supervision. This represents the primary point of data collection, where interaction logs are produced in real-time and stored for analysis [11][12].
5. Evaluation: The performance of the artefact is evaluated against the defined objectives. This includes:
    - **Technical Evaluation:** Testing the validity/accuracy and precision of data mining models in identifying patterns of development [4].
    - **Usability Evaluation:** Obtaining parental feedback regarding the clarity of reports and ease of use, ensuring the system meets accessibility standards [4].
6. Communication: The final phase of the whole process is the documentation of the whole process, including efficiency of data mining models, developed and effectiveness of the web application through the final dissertation and project reports to be presented to the Faculty of Computing.

4.3 Data Collection Methods

In order to assist in a variety of methods to provide the best coverage of toddler development this study utilized a dual method for data collection.

Examples:

- Primary data collection will be facilitated through the web application's user interface. Parents will be prompted to log specific milestones based on guided physical activities, such as their child's success in motor tasks or social mimicry [11]. This captures real-time, domestic data that traditional clinical visits might miss, providing a more longitudinal view of the child's daily progress [12].
- Secondary data will be sourced from existing clinical and educational repositories. Datasets such as the National Longitudinal Studies of Kindergarten Children [12] and clinical MRI growth records [10] will be used to train the machine learning model. This ensures the AI has a solid foundation of "ground truth" data before it begins classifying new parent-logged entries [4][13].

4.4 Data Preprocessing / Preparation

Interaction data in its original form includes noise or incomplete sessions. Data will go through the following processes:

- Cleaning of Data: This phase involves identifying and deleting sessions that are considered “noisy,” such as those that were accidentally closed, contain duplicate entries, or show zero interaction time. This step is critical to ensure that the Hybrid Machine Learning model is not trained on corrupted or irrelevant parental logs [11].
- Extraction of Features: Raw log files and timestamped activity reports are converted into meaningful features. These include the average amount of time parents spend on guided physical tasks, the completion rate of offline milestones, and the linguistic accuracy reported during social engagement sessions [4].
- Normalization: All feature values are normalized to account for the vast developmental differences between toddlers of different ages (e.g., 12 months vs. 36 months). This process allows for an age-neutral categorization of the data mining training sets, ensuring that the classification of "At Risk" or "On Track" is developmentally appropriate for each specific child [4][12].

4.5 Model Development / Implementation

The back end is based on the MERN Stack with a Python-driven analytics engine.

- Algorithms: The project implements classification models, specifically Random Forest (RF) and Support Vector Machines (SVM), to categorize a child's developmental status as "On Track," "Needs Monitoring," or "At Risk" based on features derived from parental inputs [5].
- Hybrid Models: To enhance diagnostic precision, a Hybrid model will be developed as proposed by Sokkhey & Okazaki. This approach amalgamates multiple algorithms to improve the predictive accuracy of identifying developmental delays in the early-age phenotype [4[5].
- System Components: The architecture features a React-based dashboard for parent-child activity logging, a Node.js API for secure data management in MongoDB, and a Python Flask/FastAPI server dedicated to the real-time execution of machine learning models [8][11].

4.6 Evaluation

In view of the above, the effectiveness of the solution will be judged from two major standpoints:

- Model Performances: The performances of the data mining algorithms will be measured by a Confusion Matrix, calculating metrics such as Accuracy, Precision, Recall, and F1-score [5]. This ensures that the system actually identifies developmental patterns.

- Usability of System: The interface of the application will be tested, with parents supervising, based on the work of Gibson et al., for accessibility and Engagement by toddlers in the age group of 1-3 years [4].

4.7 Ethical considerations

The study requires a high ethical code, as the participants will be the vulnerable age group, comprised of young children:

- Ethics and Consent: Explicit digital consent will be obtained from parents/guardians before the start of data collection.

- Privacy of Data: All data would be anonymized using unique IDs to mask personal information so that no child data could be identified.

- No Medical Diagnosis: It is not presented as a final medical diagnosis, but rather a "screening and monitoring tool". The software shall further provide a disclaimer that advises parents to consult healthcare professionals for clinical assessments [6].**

4.8 Limitations of the study

Although this research is intended to offer a sound framework for toddler development evaluation, some internal limitations and flaws are to be recognized:

- Sample Size and Generalizability: The pilot stage may feature a small participant pool. Since the accuracy of the Hybrid Random Forest classifier [4] depends on data volume, a limited sample size increases the risk of "overfitting," which may hinder the model's ability to generalize across diverse socio-economic backgrounds [12].
- Reporting and Interaction Bias: Reliance on parental logging introduces "Subjective Reporting Bias" linked to Maternal Self-Efficacy [2]. It remains challenging to distinguish between a toddler's autonomous developmental choice and "Parentally Assisted" interactions where a caregiver may have prompted the child [11].
- Hardware and Technical Variability: Variations in device specifications, such as CPU speed and screen responsiveness, can affect data accuracy [3]. Technical latency on older devices within the MERN stack interface may lead to improper timestamping of milestones, impacting the "Interaction Log" [5][8].
- Engagement and Behavioral Instability: Unlike controlled clinical assessments [13], home-based logging is subject to environmental distractions like noise or lighting. A toddler's fluctuating temperament and focus can lead to data entries that do not accurately represent their true developmental milestones [1, 10].
- Internet Connectivity and Accessibility: As a web-based system, stable connectivity is required for real-time synchronization. Latency in regions with poor internet access can disrupt the interactive dashboard and lead to incomplete data entries or data loss [4].
- Data Accuracy and Professional Diagnosis: The system is a screening tool, not a diagnostic replacement. While the Machine Learning model [4] identifies risks, complex disorders like Autism [6] still require specialized longitudinal assessment by professionals for a formal medical diagnosis [10][13].

5.Timeline

Phase	Activity	Duration
Phase 1	**Literature Review & Requirement Analysis:** Conducting an in-depth study of previous research and finalizing functional requirements for the parent-mediated tool.	Jan 2026 - March 2026
Phase 2	**System Design & Development:** Designing the system architecture and developing the MERN stack application with the Python analytics engine.	Apr 2026 - June 2026
Phase 3	**Data Collection & Model Training:** Deploying the prototype for parental logging and training the Hybrid Random Forest models.	July 2026 - Sept 2026
Phase 4	**Evaluation & Thesis Writing:** Technical and usability evaluation, followed by the final documentation of results.	Oct 2026 - Nov 2026
Phase 5	**Final Submission:** Final review of the project and submission of the dissertation.	Dec 2026

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